

SISTEMA DI RACCOMANDAZIONE



Prodotti cosmetici su similarità multidimensionale

*«La similarità deve riflettere
ingredienti, brand e categoria,
non solo il nome del prodotto.»*

CONTESTO

Il mercato cosmetico presenta un'elevata varietà di prodotti apparentemente simili ma con composizioni chimiche molto diverse.

I sistemi di raccomandazione tradizionali si basano spesso su descrizioni testuali o strategie di marketing, trascurando la reale formulazione del prodotto.

Questo approccio può portare a suggerimenti poco coerenti dal punto di vista funzionale e qualitativo.

OBIETTIVO

Sviluppare un sistema di raccomandazione in grado di identificare prodotti cosmetici realmente simili a un prodotto di partenza.

DATASET

Il dataset utilizzato proviene dalla piattaforma **OpenFoodFacts** ed è stato filtrato per includere esclusivamente prodotti cosmetici.

Il dataset originale è composto da circa **60.000 prodotti** e presenta dati grezzi, rumorosi e non normalizzati.

Prima dell'analisi è stata effettuata una fase approfondita di **pulizia e ristrutturazione** dei dati.

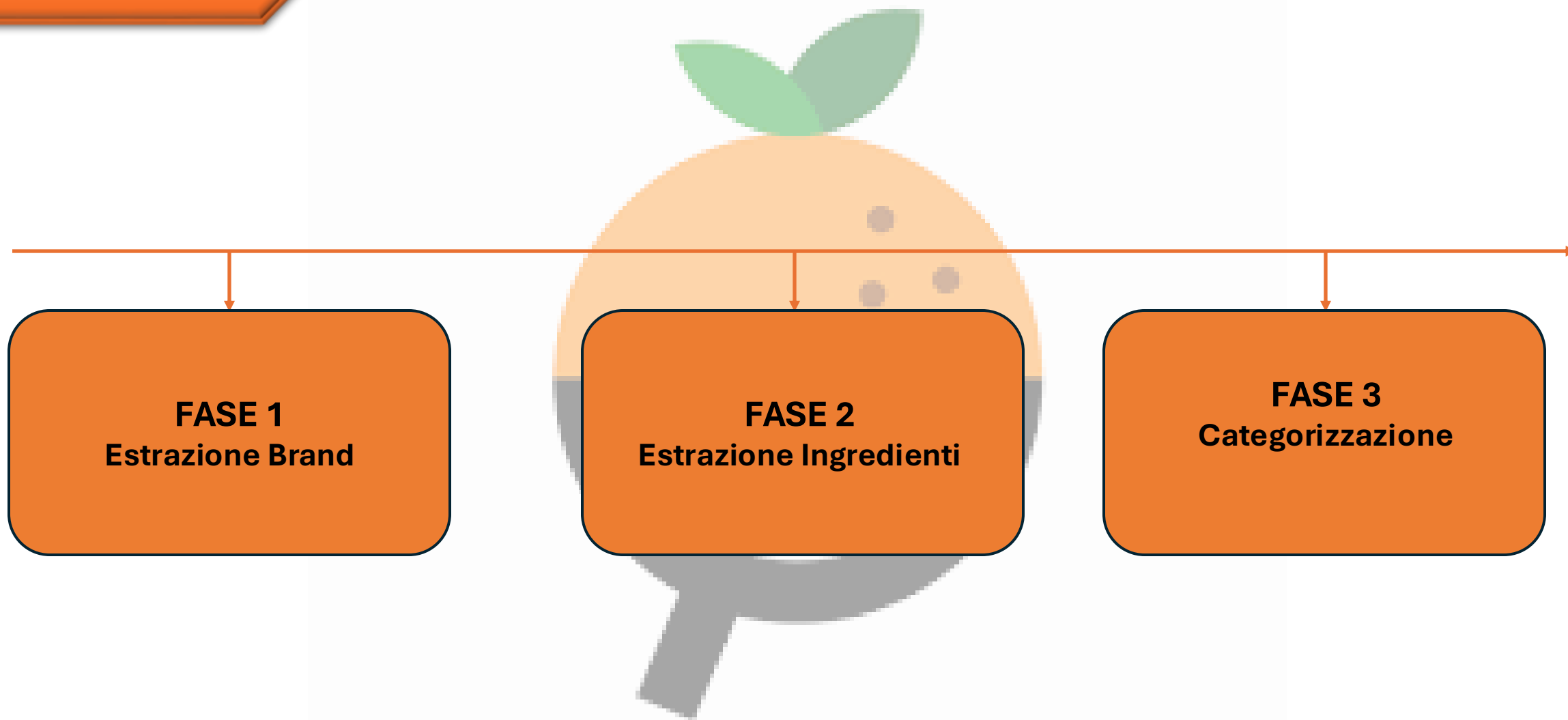


DATASET ROW

- e interpretabile

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
code	url	creator	created_t	created_d	last_modif	last_modif	last_modif	last_modif	last_modif	product_n	abbreviate	generic_n	quantity	packaging	packaging	packaging	packaging	brands	brands_tag	brands_en	categories	categories	
00000001	http://world-en.openbeautyfacts.org/product/00000001	tonyracket1	1751609682	2025-07-04	T06:14:42Z	1751609686	2025-07-04	T06:14:46Z	tonyracket1	1751609686	2025-07-04	T06:14:46Z	Arnica	ait:arnica	ait:arnica	en:italy	en:italy	italy	en:italy	italy	en:italy	italy	en:italy
00000014	http://world-en.openbeautyfacts.org/product/00000014	susslupinen-mehl-bodylab-24insect	productadd1710345659	2024-03-13	T16:00:59Z	1748277440	2025-05-26	T16:37:20Z	fitcounter1748277440	2025-05-26	T16:37:20Z	SÃ	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	
00000018	http://world-en.openbeautyfacts.org/product/00000018	chocolate-peanut-butter-protein	foodvisor1729882037	2024-10-25	T18:47:17Z	1742914310	2025-03-25	T14:51:50Z	smoothie-app1742914310	2025-03-25	T14:51:50Z	Chocolate	peanut butter	protein									
00000092	http://world-en.openbeautyfacts.org/product/00000092	airpods-appleproduct-scan-com	1638580447	2021-12-04	T01:14:07Z	1744901851	2025-04-17	T14:57:31Z	smoothie-app1744901851	2025-04-17	T14:57:31Z	Airpods1	Applexx	apple	apple	france	en:fr						
00000163	http://world-en.openbeautyfacts.org/product/00000163	smoothie-app1682257711	2023-04-23	T13:48:31Z	1727959822	2024-10-03	T12:50:22Z	fix-code-bot1743166885	2025-03-28	T13:01:25Z	Deutschland	en:germany	Germany	en:germany	Germany	en:germany	Germany	en:germany	Germany	en:germany	Germany	en:germany	
00000280	http://world-en.openbeautyfacts.org/product/00000280	date-limite-app1584379762	2020-03-16	T17:29:22Z	1744884567	2025-04-17	T10:09:27Z	smoothie-app1744884567	2025-04-17	T10:09:27Z	Conserve	conserve	Conserve	en:france	en:france	France	en:france	France	en:france	France	en:france	France	
00000311	http://world-en.openbeautyfacts.org/product/00000311	huile-vegetale-amande-douce-aroma-zone	openfoodfacts-contributors1518297016	2018-02-10	T21:10:16Z	1727959827	2024-10-03	T12:50:27Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	Huile	vÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	
00000315	http://world-en.openbeautyfacts.org/product/00000315	colgate-sensitive-plus-made-in-india-oat-haus	foodvisor1656273028	2022-06-26	T19:50:28Z	1736518675	2025-01-10	T14:17:55Z	teolemon1743166980	2025-03-28	T13:03:00Z	Colgate	Sensitive	Plus									
00000320	http://world-en.openbeautyfacts.org/product/00000320	body-earth-inc	openfoodfacts-contributors1612887819	2021-02-09	T16:23:39Z	1747473474	2025-05-17	T09:17:54Z	teolemon1747473474	2025-05-17	T09:17:54Z	Body & Earth, Inc	Body & Earth	Inc									
00000347	http://world-en.openbeautyfacts.org/product/00000347	quercetin-zinc-vitamin-c-vitamin-d-elderberry-natra	couticals-q-defend	foodvisor1714427298	2024-04-29	T21:48:18Z	1740528027	2025-02-26	T00:00:27Z	waistline-app1743166886	2025-03-28	T13:01:26Z	Quercetin	Zinc	Vitamin C	Vitamin D	Elderberry	Natracouticals	Q-Defend				
00000423	http://world-en.openbeautyfacts.org/product/00000423	desmond	d1744616417	2025-04-14	T07:40:17Z	1748671427	2025-05-31	T06:03:47Z	smoothie-app1748671427	2025-05-31	T06:03:47Z	en:united-kingdom	en:united-kingdom	United Kingdom	en:united-kingdom	United Kingdom	en:united-kingdom	United Kingdom	en:united-kingdom	United Kingdom	en:united-kingdom	United Kingdom	
00000493	http://world-en.openbeautyfacts.org/product/00000493	savon-vegetal-illuminateur-aroma-zone	openfoodfacts-contributors1565650963	2019-08-12	T23:02:43Z	1727959829	2024-10-03	T12:50:29Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	savon	vegetal	illuminateur									
00000511	http://world-en.openbeautyfacts.org/product/00000511	pure-collagene-marin-null	product-scan-com1632331280	2021-09-22	T17:21:20Z	1727959748	2024-10-03	T12:49:08Z	fix-code-bot1743168706	2025-03-28	T13:31:46Z	pure collagen	Ã	ne marin	120								
00000561	http://world-en.openbeautyfacts.org/product/00000561	huile-vegetale-coco-aroma-zone	openfoodfacts-contributors1518297122	2018-02-10	T21:12:02Z	1727959832	2024-10-03	T12:50:32Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	Huile	vÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	gÃ	
00000563	http://world-en.openbeautyfacts.org/product/00000563	beaute-des-pieds-baume-reparateur-lavande-bio-yves-rocher	angela23751466793656	2016-06-24	T18:40:56Z	1727959832	2024-10-03	T12:50:32Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z												
00000571	http://world-en.openbeautyfacts.org/product/00000571	openfoodfacts-contributors1713086550	2024-04-14	T09:22:30Z	1727959832	2024-10-03	T12:50:32Z	fix-code-bot1727959832	2024-10-03	T12:50:32Z	en:france	en:france	France	en:france	France	Olea europaea	fruit						
00000587	http://world-en.openbeautyfacts.org/product/00000587	macerat-huileux-lys-aroma-zone	openfoodfacts-contributors1518296822	2018-02-10	T21:07:02Z	1727959833	2024-10-03	T12:50:33Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	MacÃ	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	
00000592	http://world-en.openbeautyfacts.org/product/00000592	smoothie-app1744209612	2025-04-09	T14:40:12Z	1744209676	2025-04-09	T14:41:16Z	smoothie-app1744209676	2025-04-09	T14:41:16Z	en:south-africa	en:south-africa	South Africa	en:south-africa	South Africa	en:south-africa	South Africa	en:south-africa	South Africa	en:south-africa	South Africa	en:south-africa	
00000599	http://world-en.openbeautyfacts.org/product/00000599	smoothie-app1740311434	2025-02-23	T11:50:34Z	1740311587	2025-02-23	T11:53:07Z	smoothie-app1740311587	2025-02-23	T11:53:07Z	en:italy	en:italy	italy	en:italy	italy	en:italy	italy	en:italy	italy	en:italy	italy	en:italy	
00000659	http://world-en.openbeautyfacts.org/product/00000659	poires-chocolat-noir	kiliweb1558891084	2019-05-26	T17:18:04Z	1747809647	2025-05-21	T06:40:47Z	smoothie-app1747809647	2025-05-21	T06:40:47Z	Poires	chocolat	noir	en:france	en:france	France	en:france	France	en:france	France	en:france	
00000695	http://world-en.openbeautyfacts.org/product/00000695	ole-ole-aloe-loovara	openfoodfacts-contributors1601906868	2020-10-05	T14:07:48Z	1727959748	2024-10-03	T12:49:08Z	fix-code-bot1743166885	2025-03-28	T13:01:25Z	oÃ	oÃ	oÃ	oÃ	oÃ	oÃ	oÃ	oÃ	oÃ	oÃ	oÃ	
00000735	http://world-en.openbeautyfacts.org/product/00000735	creme-pour-barbe-aliver	openfoodfacts-contributors1542194248	2018-11-14	T11:17:28Z	1727959852	2024-10-03	T12:50:52Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	CrÃ	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	Ã	
00000759	http://world-en.openbeautyfacts.org/product/00000759	gel-nail-polish-myleepauline	m1685475941	2023-05-30	T19:45:41Z	1727959852	2024-10-03	T12:50:52Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	Gel	nail	polish	10ml	0.33	mylee	en:mylee	mylee	en:mylee	mylee	en:mylee	
00000834	http://world-en.openbeautyfacts.org/product/00000834	openfoodfacts-contributors1532180415	2018-07-21	T13:40:15Z	1728653863	2024-10-11	T13:37:43Z	fix-code-bot1728653863	2024-10-11	T13:37:43Z	en:france	en:france	France	en:france	France	en:france	France	en:france	France	en:france	France	en:france	
00000859	http://world-en.openbeautvfacts.org/product/00000859	base-lavante-aroma-zone	openfoodfacts-contributors1518296658	2018-02-10	T21:04:18Z	1727959834	2024-10-03	T12:50:34Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	Base	lavante	Base	lav								

PULIZIA



PULIZIA BRAND

- Estrazione degli **attributi chiave** (codice,nome, brand)
- Rimozione di **valori nulli** e record inconsistenti
- **Normalizzazione** dei formati dati
- Gestione prodotti **multi-brand**
- Assegnazione **segmento di mercato**

PRIMA

code	product_name	brand_name
9896225760529	NaN	NaN
9900172000505	NaN	NaN
9903232234964	NaN	NaN
9905907038495	NaN	NaN
9906651473495	mango	NaN
9909205010818	Yardley London Morning Dew	Yardley
9909452841586	NaN	NaN
9909684334245	NaN	NaN
9909848264562	NaN	NaN

DOPO

code	product_name	brand_name	brand_segment
5901887010371	eye make up remover	ziaja	middle
5901887010173	crema facial pepino	ziaja	middle
5901887001348	gel limpiador de aceite de oliva	ziaja	middle
5901887016953	crema facial ultra ligera aceite de oliva	ziaja	middle
5901887008460	hand cream	ziaja	middle
5901887006916	sensitive soothing day cream	ziaja	middle
3700436313589	bush magic	zingus	mass_market
7596211001033	crema hidratante dia fps 15	zoah	mass_market
7596211000203	crema corporal para la celulitis	zoah	mass_market
4770161103723	balksvuju gyslociu luobeles	zolynelis	mass_market

PULIZIA INGREDIENTI

- Analisi della **composizione** per ogni prodotto
- Estrazione dei singoli ingredienti (inci_name)
- Eliminazione valori nulli
- Creazione di rappresentazioni testuali unificate
- **Normalizzazione** formato dati
- Creazione colonna **is_functional**: per identificare i soli ingredienti con benefici
- Creazione colonna **ingredient_id**: ogni ingrediente ha un proprio identificativo

PRIMA

	code	ingredients_text	ingredients_tags	ingredients_analysis_tags
0	00000001	NaN	NaN	NaN
1	00000014	CARATTERISTICHE PRODOTTO: gel a base di acqua ...	en:caratteristiche- prodotto,en:mentolo-e-con-o...	en:palm-oil-content-unknown,en:vegan- status-un...
2	00000018	NaN	NaN	NaN
3	00000092	NaN	NaN	NaN
4	00000163	NaN	NaN	NaN
5	00000280	NaN	NaN	NaN
6	00000311	Prunus amygdalus dulcis oil	fr:prunus-amygdalus-dulcis-oil	en:palm-oil-content-unknown,en:vegan- status-un...
7	00000315	NaN	NaN	NaN
8	00000320	NaN	NaN	NaN
9	00000347	NaN	NaN	NaN

DOPO

	code	product_name	ingredient_id	inci_name	is_functional
0	00000014	Süßlupinen Mehl	1	Water	1
1	00000014	Süßlupinen Mehl	2	Buriti Oil	1
2	00000014	Süßlupinen Mehl	3	Arnica Montana	1
3	00000014	Süßlupinen Mehl	4	Menthol	1
4	00000014	Süßlupinen Mehl	5	Avoid Contact With Eyes	1
5	00000014	Süßlupinen Mehl	6	Mucous	1
6	00000014	Süßlupinen Mehl	7	Tocopherol	1
7	00000014	Süßlupinen Mehl	8	Oil	1
8	00000014	Süßlupinen Mehl	9	Phenoxyethanol R Um Sorbate	0
9	00000014	Süßlupinen Mehl	10	Silicones	1

RISULTATO



Una lista di ingredienti
per ogni prodotto

ASSEGNAZIONE CATEGORIA

- Inserimento della **macro-categoria** per ogni tabella
- **Mappatura** prodotti-categorie (Deodorants, HairCare, Face Care, ecc.)
- Verifica della **coerenza** categoriale
- Creazione di **dataset finale integrato**

BRAND

code	product_name	brand_name	brand_segment	macro_category
5901887010371	eye make up remover	ziaja	middle	Makeup
5901887010173	crema facial pepino	ziaja	middle	Face_Care
5901887001348	gel limpiador de aceite de oliva	ziaja	middle	Oils
5901887016953	crema facial ultra ligera aceite de oliva	ziaja	middle	Face_Care
5901887008460	hand cream	ziaja	middle	Face_Care
5901887006916	sensitive soothing day cream	ziaja	middle	Face_Care
3700436313589	bush magic	zingus	mass_market	Other
7596211001033	crema hidratante dia fps 15	zoah	mass_market	Face_Care
7596211000203	crema corporal para la celulitis	zoah	mass_market	Body_Care
4770161103723	balksvuju gyslociu luobeles	zolynelis	mass_market	Other
8809576260489	zombie beauty	zombie beauty	mass_market	Other

INGREDIENTI

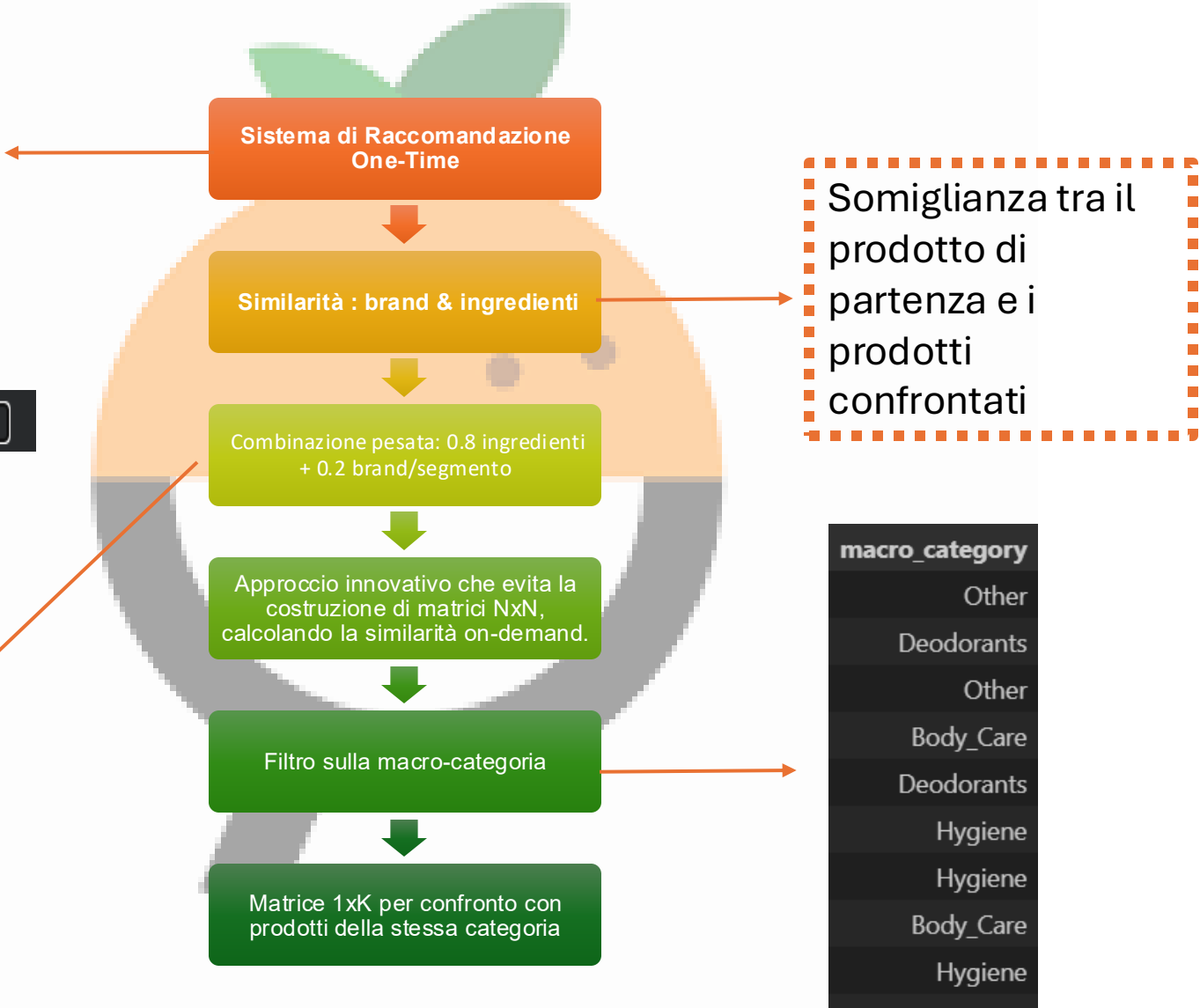
code	product_name	macro_category	ingredients
96121214	Powder soft anti perspirant	Other	Alpha-Isomethyl Ionone, Aluminum Zirconium Tet...
96125694	REXONA Déodorant Homme Bille Anti Transpirant ...	Deodorants	Alpha-Isomethyl Ionone, Aluminum Zirconium Tet...
96125748	Motionsense active shield	Other	Alpha-Isomethyl Ionone, Aluminum Chlorohydrate...
96130339	Dove zero original 50ml	Body_Care	Alpha-Isomethyl Ionone, Benzyl Alcohol, Citric...
96132289	Axe Déodorant Anti Bactérien You Spray 100ml	Deodorants	Alcohol Denat, Alpha-Isomethyl Ionone, Butane,...
96132876	Monsavon Déodorant Femme Spray Compressé Fleur...	Hygiene	Alpha-Isomethyl Ionone, Benzyl Alcohol, Bht, B...
96132906	Monsavon Déodorant Femme Spray Compressé Grena...	Hygiene	Alpha-Isomethyl Ionone, Benzyl Alcohol, Bht, B...
96134696	DOVE Déodorant Femme Spray Compressé Pierre d'...	Body_Care	Alpha-Isomethyl Ionone, Benzyl Alcohol, Benzyl...
96148402	Monsavon Déodorant Femme Spray Compressé Vanil...	Hygiene	Benzyl Alcohol, Bht, Butane, Butyrospermum Par...
96183281	sensodyne	Other	Aroma, Carbomer, Cocamidopropyl Betaine, Die 7...
96190807	Clay Pomade	Other	Bentonite, Ceteareth-20, Ethylhexyl Palmitate,...
0150153491	nutri-extrême richesse	Other	Alpha-Isomethyl Ionone, Benzyl Alcohol, Benzyl...

SIMILARITA'

Inserendo il codice del prodotto, il sistema suggerirà i prodotti più simili appartenenti alla stessa categoria

Inserisci il codice del prodotto:

	code	ing_sim	brand_sim
0	3094905000033	0.916667	0.7
1	5054563155658	0.916667	1.0
2	5054563006592	0.916667	1.0
3	5054563109743	0.733333	1.0
4	5054563010278	0.666667	1.0



SIMILARITA' INGREDIENTI

1. Ingredienti di ciascun prodotto vengono aggregati e trasformati:

- colonna con tutti gli ingredienti
- colonna con lista di attivi funzionali

Ogni prodotto descritto come un vettore di frequenze degli ingredienti

2. Calcolata similarità Jaccard tra prodotto ricercato e candidati

Si ottiene una misura quantitativa della somiglianza tra i prodotti

RISULTATO

Una matrice 1xK con i soli prodotti della medesima categoria e con identici attivi funzionali

code	product_name	macro_category	ingredients	ingredients_list	ing_sim	is_query	common_ingr
9327693006920	Sukin haircare	Hair_Care	Aloe Barbadensis Leaf Juice, Arctium Lappa Roo...	[glycerin, phenoxyethanol, limonene, arctium l...	1.000000	True	[glycerin, phenoxyethanol, limonene, arctium l...
9327693009020	Sukin Colour Care Shampoo	Hair_Care	Benzyl Alcohol, Betula Alba Leaf Extract, Citr...	[glycerin, betula alba leaf extract, phenoxyet...	0.473684	False	[glycerin, phenoxyethanol, limonene, tocophero...
9327693007071	Hydrating Conditioner	Hair_Care	Aloe Barbadensis Leaf Juice, Benzyl Alcohol, B...	[glycerin, isopropyl alcohol, phenoxyethanol, ...	0.388889	False	[glycerin, phenoxyethanol, limonene, tocophero...
4005808677702	Shampooing soin douceur hydra douceur	Hair_Care	Aloe Barbadensis Leaf Juice, Alpha-Isomethyl I...	[glycerin, alpha-isomethyl ionone, limonene, g...	0.304348	False	[glycerin, limonene, cocamidopropyl betaine, o...
08228895	shampoo	Hair_Care	Aloe Barbadensis Leaf Juice, Benzyl Benzoate, ...	[glycerin, benzyl benzoate, phenoxyethanol, gl...	0.304348	False	[glycerin, phenoxyethanol, tocopherol, cocamid...

SIMILARITA' BRAND

1. Verificata appartenenza allo stesso brand

Prodotti appartenenti allo stesso brand ricevono il punteggio massimo: 1.0

2. Calcolo similarità coseno considerando il segmento di mercato del brand

Se il segmento è lo stesso, al punteggio viene aggiunto un incremento di 0.5 per rafforzare l'affinità tra prodotti simili.

CALCOLO

Attraverso una funzione deterministica basata su regole di business, senza l'utilizzo di modelli di machine learning.

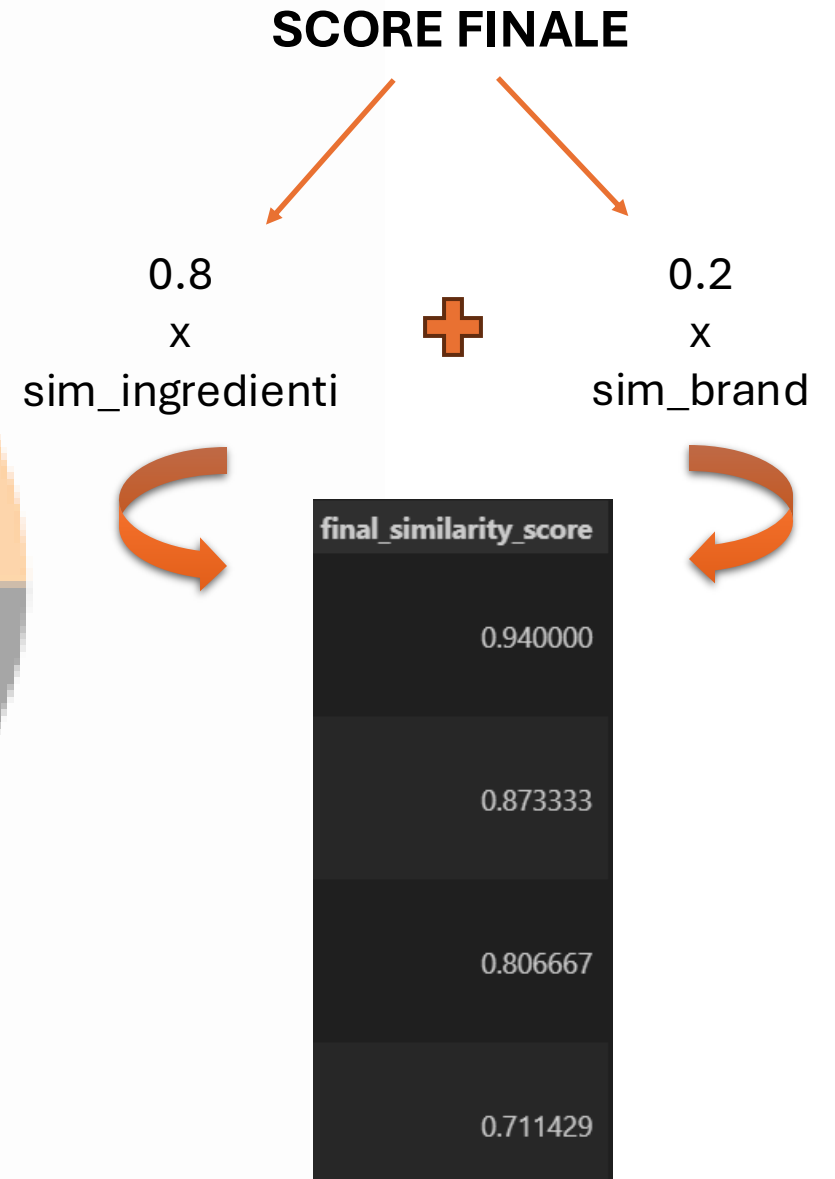
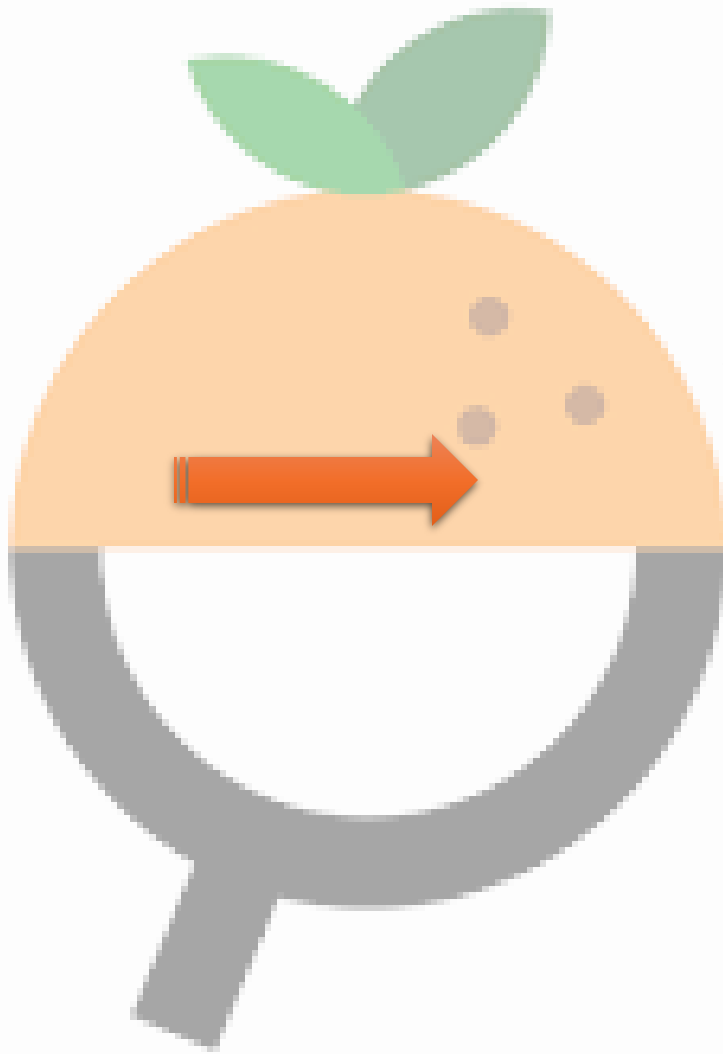
RISULTATO

Una matrice 1xK che confronta il prodotto di partenza con gli altri prodotti della **stessa categoria** producendo un risultato chiaro ed interpretabile.

	code	query_product_name	query_brand_name	product_name	brand_name	brand_segment	macro_category	similarity
7237	3245678612491	deodorant power 24h	cosmia	deodorant energy 24h	cosmia	middle	Deodorants	1.0
7235	3596710509287	deodorant power 24h	cosmia	deodorant	cosmia	middle	Deodorants	1.0
7247	3245678612682	deodorant power 24h	cosmia	deodorant voluptuous time	cosmia	middle	Deodorants	1.0
7254	3245678612415	deodorant power 24h	cosmia	deodorant instant voluptueux	cosmia	middle	Deodorants	1.0
7260	3245678612743	deodorant power 24h	cosmia	deodorant fleurs blanches 24h	cosmia	middle	Deodorants	1.0
7261	3245678612729	deodorant power 24h	cosmia	deodorant vanille	cosmia	middle	Deodorants	1.0

COMBINAZIONE

Il risultato è **perfettamente allineato** per prodotto, permettendo combinazione lineare diretta.



OUTPUT

Codice	Nome	Ingredienti	Codice	Nome	Brand	Segmento
query_product_id	query_product_name	ingredients_query	recommended_product_id	product_name	brand_name	brand_segment
3094905000033	repare protege	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563006592	sensodyne repair protect	sensodyne	mass_market
3094905000033	repare protege	Aroma, Calcium Sodium Phosphosilicate, Carbome...	8901571010844	sensodyne repair protect	sensodyne	mass_market
3094905000033	repare protege	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563155658	repair protect	sensodyne	mass_market
3094905000033	repare protege	Aroma, Calcium Sodium Phosphosilicate, Carbome...	05050499	sensodyne sensibilite gengives	gsk	mass_market
3094905000033	repare protege	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563010278	rapide action extra fresh	sensodyne	mass_market
3094905000033	repare protege	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563109743	sensodyne sensibilidade e gengivas	sensodyne	mass_market
3094905000033	repare protege	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563024558	rapide action	sensodyne	mass_market

Prodotto cercato

Prodotti raccomandati

ESEMPIO OUTPUT 1

rank	query_product_id	query_product_name	ingredients_query	recommended_product_id	product_name	brand_name	brand_segment	common_ingredients	num_common_ingredients	final_similarity_score
1	8500033101160	tula gentle daily facial cleanser	Alpha-Isomethyl Ionone, Ascorbyl Palmitate, Be...	0850003310312	the cult classic purifying face cleanser	tula	luxury	[disteareth-75 ipdi, sodium lauryl sulfoacetat...	32	0.675676
2	8500033101160	tula gentle daily facial cleanser	Alpha-Isomethyl Ionone, Ascorbyl Palmitate, Be...	0850014698355	24 7 moisture hydrating day amp night cream	tula	luxury	[disteareth-75 ipdi, sodium lauryl sulfoacetat...	22	0.466667
3	8500033101160	tula gentle daily facial cleanser	Alpha-Isomethyl Ionone, Ascorbyl Palmitate, Be...	0809280149078	travel size soy face cleanser	fresh	luxury	[tocopheryl acetate, pentylene glycol, curcuma...	17	0.300000
4	8500033101160	tula gentle daily facial cleanser	Alpha-Isomethyl Ionone, Ascorbyl Palmitate, Be...	3378872098195	fleur de coton creme lavante mains	sephora	luxury	[tocopheryl acetate, ethylhexylglycerin, oil, ...	8	0.296098
5	8500033101160	tula gentle daily facial cleanser	Alpha-Isomethyl Ionone, Ascorbyl Palmitate, Be...	6909122263956	purity made simple one step facial cleanser	philosophy	luxury	[pentylene glycol, caprylyl glycol, ethylhexyl...	9	0.282222
6	8500033101160	tula gentle daily facial cleanser	Alpha-Isomethyl Ionone, Ascorbyl Palmitate, Be...	3614229839456	purity made simple one step facial cleanser wi...	philosophy	luxury	[tocopheryl acetate, caprylyl glycol, ethylhex...	10	0.282222

Ordine
decrescente

Top-K raccomandazioni ordinate per score finale

ESEMPIO OUTPUT 2

rank	query_product_id	query_product_name	ingredients_query	recommended_product_id	product_name	brand_name	brand_segment	common_ingredients	num_common_ingredients	final_similarity_score
1	8901571010844	sensodyne repair protect	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563006592	sensodyne repair protect	sensodyne	mass_market	[titanium dioxide, sodium saccharin, sodium me...	12	0.933333
2	8901571010844	sensodyne repair protect	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563155658	repair protect	sensodyne	mass_market	[titanium dioxide, sodium saccharin, sodium me...	11	0.933333
3	8901571010844	sensodyne repair protect	Aroma, Calcium Sodium Phosphosilicate, Carbome...	3094905000033	repare protege	glaxosmithkline	mass_market	[titanium dioxide, sodium saccharin, sodium me...	11	0.873333
4	8901571010844	sensodyne repair protect	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563109743	sensodyne sensibilidade e gengivas	sensodyne	mass_market	[titanium dioxide, sodium saccharin, sodium me...	11	0.786667
5	8901571010844	sensodyne repair protect	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563010278	rapide action extra fresh	sensodyne	mass_market	[titanium dioxide, sodium saccharin, cocamidop...	11	0.733333
6	8901571010844	sensodyne repair protect	Aroma, Calcium Sodium Phosphosilicate, Carbome...	5054563024558	rapide action	sensodyne	mass_market	[titanium dioxide, sodium saccharin, cocamidop...	10	0.680000

Visualizzazione **score finale** ingredienti e brand

ESEMPIO OUTPUT 3

rank	query_product_id	query_product_name	ingredients_query	recommended_product_id	product_name	brand_name	brand_segment	common_ingredients	num_common_ingredients	final_similarity_score
1	3600522862710	elseve arginine resist x3 formule 2 en 1	2-Oleamido-1, 3- Octadecanediol, Arginine, Carb...	3610340010248	elseve arginine resist x3	l'oreal	mass_market	[hydroxypropyltrimonium hydrolyzed wheat prote...	21	1.000000
2	3600522862710	elseve arginine resist x3 formule 2 en 1	2-Oleamido-1, 3- Octadecanediol, Arginine, Carb...	3600523169900	elseve arginine resist x3 format xxl	l'oreal	mass_market	[hydroxypropyltrimonium hydrolyzed wheat prote...	21	1.000000
3	3600522862710	elseve arginine resist x3 formule 2 en 1	2-Oleamido-1, 3- Octadecanediol, Arginine, Carb...	3600522383703	elseve arginine resist x3 maxi format	l'oreal	mass_market	[hydroxypropyltrimonium hydrolyzed wheat prote...	21	1.000000
4	3600522862710	elseve arginine resist x3 formule 2 en 1	2-Oleamido-1, 3- Octadecanediol, Arginine, Carb...	3600522836872	elseve total repair 5	l'oreal	mass_market	[hydroxypropyltrimonium hydrolyzed wheat prote...	21	0.653333
5	3600522862710	elseve arginine resist x3 formule 2 en 1	2-Oleamido-1, 3- Octadecanediol, Arginine, Carb...	2000000000050	fructis classic	l'oreal	mass_market	[hydroxypropyltrimonium hydrolyzed wheat prote...	13	0.569231
6	3600522862710	elseve arginine resist x3 formule 2 en 1	2-Oleamido-1, 3- Octadecanediol, Arginine, Carb...	3600550166620	expert reparation	franck provost	mass_market	[hydroxypropyltrimonium hydrolyzed wheat prote...	17	0.540000

Lista e numero **ingredienti comuni** tra
prodotto ricercato e prodotti raccomandati



GRAZIE PER L'ATTENZIONE

Corso: Data & AI Analyst
Tutor: Pier Giorgio Poloni

